Wiring Diagram v003 (WIP)

Dustie Mk2

aka 'The VHS Wood CNC'

NEMA 6-15
240VAC
240v R, S, PE
Strain Relief Grommet
DIN Rail Terminal Block
R
S
PE
MCB 2P 15A TYPE C
AC EMI Filter
L
N
GND
PSU · 120 W · 24 V
Mesa 7i76EU
E Stop
with indicator lamp
EMERGENCY STOP
NC
NO
Machine Stop
NO Momentary button
A1
1/L1
3/L2
5/L3
13 NO
A2
AC Contactor
2/T1
4/T2
6/T3
14 NO
Terminal Blocks
6P Blue (3× blocks), 6P Grey (3× blocks), 7P PE (4× blocks)
PSU · 350 W · 48 V
Y1 Stepper, Y2 Stepper
PSU · 350 W · 48 V
X Stepper, Z Stepper
PSU · 350 W · 24 V
Mesa 7i76EU, Rotary Axis A, Field Power
V+
V+
V+
V-
V-
V-
PE
N
L
Lenovo PC · LinuxCNC
← DVI to HDMI →
USB Extension
Lenovo
Ethernet
HY-DIV268N-5A
RJ45 ETH
Mesa 7i76EU
FRAME GROUND
Inside Cabinet
Outside Cabinet
Umbilical
U
V
W
GND
Wall Mounted
Variable Frequency Drive
R
S
T
P+
PR
U
V
W
GND
Water
Relay-controlled
Spindle Coolant Pump
Signal
Mains AC
DB25
Breakout Box 1
Static Platform
aka 'The Table'
Multiple Connectors (4,2,2,2)
Breakout Box 2
Y Platform
aka 'The Gantry'
Through-wire (4,4)
Connectors (2,2)
Connector (2)
Breakout Box 3
X Platform
aka 'The Toolhead'
Multiple Connectors (4,2,2)
Y1 Drag Chain
X Drag Chain
Power and Signal bundles separated in drag chain
DB25 (23 signals)
DB25 (18 signals)
DB9 (7 signals)
Stepper Y1
Stepper Y2
Stepper X
Stepper Z
PNP
NO & NC
Touch Plate
Static Platform Sensors
Endstops: Y+, Y-
Home Sensors: Y1, Y2
Touch Plate
Y Platform Sensors
Endstops: X+, X-
Home Sensors: X
Z Platform Sensors
Endstops: Z+, Z-
Home Sensors: Z

WIP

E-Stop Loop improvements
Latching Machine Start Circuit
PC Power 'Relay'

WIP

Rotary Axis A/B